
13. ARTERIAL HYPERTENSION – Definition & Kinds (15 points)

1. Arterial hypertension is a **persistent elevation of systemic arterial blood pressure**.
2. Usually defined as BP **≥140/90 mmHg** on repeated measurements.
3. It is a **chronic cardiovascular disorder**.
4. Results from imbalance between **cardiac output and peripheral resistance**.
5. May be **asymptomatic for long periods**.
6. Leads to **target organ damage** (heart, brain, kidneys, eyes).
7. **Primary (essential) hypertension** – no identifiable cause.
8. Accounts for **90–95%** of cases.
9. **Secondary hypertension** – due to known causes.
10. Isolated systolic hypertension – systolic ↑, diastolic normal.
11. Malignant hypertension – rapid BP rise with organ damage.
12. Can be acute or chronic in presentation.
13. Strong genetic predisposition.
14. Lifestyle factors play major role.
15. Major risk factor for cardiovascular mortality.

14. CLASSIFICATION OF HYPERTENSIVE CONDITIONS (15 points)

1. Classification based on **etiology, severity, and course**.
2. **Primary (essential) hypertension** – multifactorial.
3. **Secondary hypertension** – identifiable cause present.
4. Renal hypertension – parenchymal or renovascular.
5. Endocrine hypertension – hormonal excess.
6. Neurogenic hypertension – CNS causes.
7. Vascular hypertension – coarctation of aorta.
8. Drug-induced hypertension – steroids, OCPs, NSAIDs.
9. Mild hypertension: 140–159 / 90–99 mmHg.
10. Moderate hypertension: 160–179 / 100–109 mmHg.
11. Severe hypertension: ≥180 / ≥110 mmHg.
12. Benign hypertension – slow progression.

13. Malignant hypertension – rapid progression.
 14. Sustained vs labile hypertension.
 15. Classification helps in treatment planning.
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15. TYPICAL FORMS OF CHANGES IN BLOOD PRESSURE (15 points)

1. Blood pressure normally fluctuates within limits.
 2. Sustained hypertension – persistent elevation.
 3. Labile hypertension – wide BP fluctuations.
 4. Stress and emotions trigger labile BP.
 5. White-coat hypertension occurs in clinical settings.
 6. BP normal at home in white-coat HTN.
 7. Isolated systolic hypertension common in elderly.
 8. Due to reduced arterial elasticity.
 9. Orthostatic hypotension – BP fall on standing.
 10. Caused by autonomic dysfunction.
 11. Nocturnal dipping – physiological BP fall during sleep.
 12. Non-dipping pattern seen in chronic HTN.
 13. Morning surge increases CV risk.
 14. BP variability affects organ damage.
 15. BP patterns guide diagnosis and therapy.
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16. ENDOCRINE ARTERIAL HYPERTENSION – Kinds & Pathogenesis (15 points)

1. Caused by **hormonal imbalance**.
2. Considered a form of secondary hypertension.
3. Pheochromocytoma – excess catecholamines.
4. Causes intense vasoconstriction.
5. Leads to episodic or sustained hypertension.
6. Hyperaldosteronism – excess aldosterone.
7. Sodium and water retention occur.
8. Blood volume increases.
9. Hypokalemia commonly present.

10. Cushing's syndrome – excess cortisol.
 11. Cortisol increases catecholamine sensitivity.
 12. Thyrotoxicosis increases cardiac output.
 13. Produces systolic hypertension.
 14. Hypothyroidism increases peripheral resistance.
 15. Causes diastolic hypertension.
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17. RENO-PARENCHYMAL ARTERIAL HYPERTENSION – Pathogenesis (15 points)

1. Caused by diseases of renal parenchyma.
 2. Most common cause of secondary hypertension.
 3. Seen in chronic kidney diseases.
 4. Loss of functional nephrons occurs.
 5. Sodium excretion decreases.
 6. Water retention increases.
 7. Blood volume expands.
 8. Cardiac output initially increases.
 9. Renal ischemia activates RAAS.
 10. Renin secretion increases.
 11. Angiotensin II causes vasoconstriction.
 12. Aldosterone increases sodium retention.
 13. Peripheral resistance rises.
 14. Hypertension becomes sustained.
 15. Volume-dependent hypertension is characteristic.
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18. VASORENAL (RENOVASCULAR) ARTERIAL HYPERTENSION – Pathogenesis (15 points)

1. Caused by reduced blood flow to kidney.
2. Due to renal artery stenosis.
3. Most commonly due to atherosclerosis.
4. Fibromuscular dysplasia in young patients.
5. Affected kidney perceives hypotension.
6. Juxtaglomerular cells release renin.

7. Renin activates RAAS.
 8. Angiotensin I converted to angiotensin II.
 9. Angiotensin II causes vasoconstriction.
 10. Peripheral resistance increases.
 11. Aldosterone secretion increases.
 12. Sodium and water retention occur.
 13. Blood volume increases.
 14. Hypertension becomes renin-dependent.
 15. Often severe and resistant to drugs.
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19. ARTERIAL HYPOTENSION – Types & Causes (15 points)

1. Arterial hypotension is abnormally low BP.
 2. Usually BP <90/60 mmHg.
 3. Leads to reduced tissue perfusion.
 4. Physiological hypotension seen in athletes.
 5. Occurs during sleep and pregnancy.
 6. Acute hypotension occurs suddenly.
 7. Caused by hemorrhage or shock.
 8. Seen in dehydration and burns.
 9. Chronic hypotension develops slowly.
 10. Seen in endocrine disorders (Addison's).
 11. Autonomic dysfunction causes hypotension.
 12. Orthostatic hypotension on standing.
 13. Due to baroreceptor failure.
 14. Causes dizziness and syncope.
 15. Severe hypotension leads to shock.
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✅ Next options (tell me):

-  **Convert these 15-point answers into a PDF**
-  **Flowcharts + diagrams**
-  **MCQs & viva questions**

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